VHF Navigation System – General Data

1. General

The VHF Navigation (VHF NAV) system receives VHF Omni Range (VOR), Localizer (LOC) and Glide Slope (GS) signals.

2. Description

A. VHF Navigation

The VHF NAV system consists of a 200 channel VHF Navigation and a 40 channel Glide Slope receiver. The VHF NAV signals give the aircraft course navigation information received from a VOR ground station. Localizer and Glide Slope flight path deviation signals received from the instrument landing system (ILS) is then displayed on the PFD.

The VOR/LOC antenna is installed on the vertical stabilizer. A dual NAV coupler (CI–502) is installed to allow simultaneous use of two VOR/LOC receivers from a single antenna. The dual output GS antenna is installed on the nose bulkhead at FS6.

The VHF NAV system is a integral component of the KTR 2280 Multi-Mode Digital Radio (MMDR) (refer to 34-51-00, Multi-Mode Digital Radio (MMDR) – General Data).

3. Adjustment/Test

A. Function Test VHF Navigation (VOR, LOC, GS) System

1) General

These procedures are for the function test of the VHF NAV system NAV 1 and NAV 2.

2) Function Test

a) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

b) Wait several minutes for the aircraft electrical and avionics systems to initialize.

c) Ensure that MMDR 1 (F3) (Left Bus) and/or MMDR 2 (F4) (Right Bus), circuit breakers on the avionics circuit breaker panel are closed.

d) Set up VHF Navigation ramp test unit, in accordance with the instructions in the test set manual.

e) Enable a MOD TONE on the ramp test unit. Verify a tone is heard when NAV 1 and NAV 2 are selected on the audio control panel.

f) Tune NAV 1 and NAV 2 frequency to a corresponding LOC ramp test unit frequency. Verify correct LOC/GS 1 and LOC/GS 2 indication on the HSI.

g) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

B. Electrical Bonding Test – VOR/LOC Antenna

1) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

2) Rotate rudder to gain sufficient access to Access and Inspection Provision 6a panel. (Refer to 12-00-00, Servicing – General Data).

3) Remove screws and then remove access panel.

4) Verify that the electrical conductivity is consistent over the electrical bond contact area by placing a probe on the VOR/LOC antenna mounting screw. Place the second probe on exposed surface of access panel doubler. The resistance between the two points must be < 3.0 milliohms unless stated otherwise.

5) Check the electrical conductivity between several random mounting screws within the electrical bond contact area to ensure consistency.

6) If the electrical conductivity is not consistent over the electrical bond contact area check the antenna electrical bonding in accordance with 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data.

7) Place Access and Inspection Provision 6a panel onto access cavity and tighten the retaining screws.

8) Return rudder to neutral position.

C. Electrical Bonding Test – Glide Slope Antenna (DM N25–2)

1) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

2) Gain access to the Glide Slope antenna which is mounted on the forward bulkhead inside the radome nose assembly. (Refer to 53-10-02, Forward Baggage Compartment – General Data).

3) Verify that the electrical conductivity is consistent over the electrical bond contact area by placing a probe on the head of the Glide Slope mounting screw. Place the second probe on the nearest ground stud. The resistance between the two points must be < 3.0 milliohms unless stated otherwise.

4) Check the electrical conductivity between several random mounting screws within the electrical bond contact area to ensure consistency.

5) If the electrical conductivity is no consistent over the electrical bond contact area check the antenna electrical bonding in accordance with 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data.

6) Reattach radome nose assembly and tighten retaining screws. (Refer to 53-10-02, Forward Baggage Compartment – General Data).

4. Removal/Installation

The VHF Navigation system is a integral component of the KTR 2280 Multi-Mode Digital Radio (MMDR), refer to 34-51-00, Multi-Mode Digital Radio (MMDR) – General data, Para 4. for the MMDR Removal/Installation procedure.

A. Removal/Installation of the VOR/LOC Antenna (DM N4–4)

1) General

These procedures are for the removal and installation of the VOR/LOC antenna.

2) Removal

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Pull the MMDR 1 (F3) (Left Bus) and/or MMDR 2 (F4) (Right Bus) circuit breaker on the avionics circuit breaker panel and fit safety clips.

c) Carefully cut the sealant from around the base of the antenna.

d) Remove the screws (24) from the base of the antenna.

e) Carefully lift the antenna clear of the fuselage until there is access to the coaxial connection.

f) Disconnect the coaxial connector.

Note Stow the connectors of the removed components.

3) Installation

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure that the MMDR 1 (F3) (Left Bus) and/or MMDR 2 (F4) (Right Bus) circuit breaker on the avionics circuit breaker panel are pulled and fitted with safety clips.

c) Solvent clean the electrical contact bond areas immediately before the installation (refer to 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data).

d) Bring the antenna assembly close to the fuselage, connect the coaxial connector.

e) Secure the antenna assembly on fuselage with the mounting screws (24).

f) Solvent clean the exposed area of the airframe around the antenna (refer to 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data).

g) Touch up the exposed area of airframe around the antenna if required with F19 primers by brush within 24 hours of the antenna installation. Ensure that the primer re-finishing provides complete coverage of the affected area.

h) If the affected area requires paint touch up, brush coat the area with the top-coat paint system matching the existing.

i) Seal around the antenna base and dome seal the securing screws. (Refer to 20-80-00, Installation of Aircraft Antennas – General data). The aircraft will be ready to fly approximately 24 hours after sealing.

j) Check the antenna electrical bonding as per Para 3.B.

k) Remove safety clips and close MMDR 1 (F3) (Left Bus) and/or MMDR 2 (F4) (Right Bus) circuit breaker on the avionics circuit breaker panel.

l) Do a function test on the NAV system as per Para 3.A.2).

B. Removal/Installation of GS Antenna (DM N25–2)

1) General

These procedures are for the removal and installation of the GS antenna (DM N25–2).

2) Removal

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Pull the MMDR 1 (F3) (Left Bus) and/or MMDR 2 (F4) (Right Bus) circuit breaker on the avionics circuit breaker panel and fit safety clips.

c) Carefully cut the sealant from around the base of the antenna.

d) Remove the screws (6) from the base of the antenna.

e) Carefully lift the antenna clear of the fuselage until there is access to the coaxial connection.

f) Disconnect the coaxial connector.

Note Stow the connectors of the removed components.

3) Installation

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure that the MMDR 1 (F3) (Left Bus) and/or MMDR 2 (F4) (Right Bus) circuit breaker on the avionics circuit breaker panel are pulled and fitted with safety clips.

c) Solvent clean the electrical contact bond areas immediately before the installation. (Refer to 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data).

d) Bring the antenna assembly close to the fuselage, connect the coaxial connector.

e) Secure the antenna assembly on fuselage with the mounting screws (6).

f) Solvent clean the exposed area of the airframe around the antenna. (Refer to 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data).

g) Touch up the exposed area of airframe around the antenna if required with F19 primers by brush within 24 hours of the antenna installation. Ensure that the primer re-finishing provides complete coverage of the affected area.

h) If the affected area requires paint touch up, brush coat the area with the top-coat paint system matching the existing.

i) Seal around the antenna base and dome seal the securing screws. (Refer to 20-80-00, Installation of Aircraft Antennas – General data). The aircraft will be ready to fly approximately 24 hours after sealing.

j) Check the antenna electrical bonding as per Para 3.C.

k) Remove safety clips and close MMDR 1 (F3) (Left Bus) and/or MMDR 2 (F4) (Right Bus) circuit breaker on the avionics circuit breaker panel.

l) Do a function test on the NAV system as per Para 3.A.2).

5. Troubleshooting

The VHF Navigation system is a integral component of the KTR 2280 Multi-Mode Digital Radio (MMDR), refer to 34-51-00, Multi-Mode Digital Radio (MMDR) – General data, Para 5. for the MMDR Troubleshooting procedure.